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16 MARCH 1964
SUPERSEDING
MIL-R-46656
11 APRIL 1962

MILITARY SPECIFICATION

REMOVER, AIRCRAFT CANOPY, M4 ASSEMBLY

1. SCOPE

1.1 This specification covers Remover, Aircraft Canopy, M4 Assembly, loaded with Cartridge, Aircraft Canopy, Remover, M29A2, for use in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 —Ammunition and Special Weapons; General Specification for.
MIL-I-45208 —Inspection System Requirements.

STANDARDS

MILITARY

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109—Quality Assurance Terms and Definitions.
MIL-STD-414—Sampling Procedures and Tables for Inspection by Variables for Percent Defective.
MS-28741 —Hose Assembly-Detachable End Fitting, Medium Pressure.

DRAWINGS

MUNITIONS COMMAND

F-8595439—Remover, Aircraft Canopy, M4, Assembly.
C-8823715—Container, Ammunition, Fiber M418 for Remover, Aircraft Canopy, M4 with Cartridge M29A2 or M5 with Cartridge M120.
C-8823716—Box, Packing, Ammunition for Remover, Aircraft Canopy, M4 with Cartridge M29A2 or M5 with Cartridge M120.
IEL-8595439 —Index to Inspection Equipment List for Remover Aircraft Canopy, M4, Assembly.

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. Materials, parts and assemblies shall comply with the requirements

FSC 1375

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specified on the applicable drawings, referenced specifications and the following:

3.2 **Torque requirements.** Torque requirements shall be as specified in table I. No loosening or deformation shall occur upon application of the breakway torque.

TABLE I.

Component	Seating or Bottoming (inch pounds)	Breakway (inch pounds)
Cap and Outside Tube.....	500 +50	400 min.
Sleeve and Outside Tube...	500 +50	400 min.
Firing Head and Inside Tube	400 +50	300 min.

3.3 **Assembly.**

3.3.1 The boss face surface of firing pin stop shall be in complete contact with cartridge.

3.3.2 The firing head assembly shall be free to rotate 360 degrees in either direction.

3.4 **Residual magnetism.** The remover shall not deflect the indicator of the compass more than 3 degrees in either direction when passed in front of the compass at a distance of 6 inches.

3.5 **Functional.**

3.5.1 *Ballistics.* Removers loaded with accepted M29A2 Cartridges shall propel a 300 pound mass vertically and shall comply with the requirements listed in table II.

3.5.2 *Locked shut.* Removers loaded with accepted M29A2 Cartridges and temperature conditioned at minus 65 degrees and plus 160 degrees F. shall withstand the generated gas pressures with no evidence of leakage or mechanical failure (see 6.1) of exterior components when fired locked shut (see 6.3).

3.5.3 The remover shall not fail to function.

3.5.4 The remover shall function without mechanical failure (see 6.1).

3.6 **Workmanship.** The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.6.1 *Metal defects.* All components shall be free from cracks, splits, cold shuts, inclusions, porosity or any similar defect.

3.6.2 *Burr.* No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.6.3 *Foreign matter.* No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 *Quality assurance terms and definitions.* Reference shall be made to Standard MIL–STD–109 to define the quality assurance terms used.

4.1.2 *Inspection.* Inspection shall be in accordance with Specifications MIL–A–2550 and MIL–I–45208, except as specified herein.

4.2 **Inspection provisions.**

TABLE II.

Temperature	Velocity, minimum fps (see 6.4)	Peak thrust minimum pounds	Ignition delay maximum (see 6.2)
—65° to —70° F.	17	2000	150 milliseconds
65° to 75° F.	20	2800	150 milliseconds
160° to 165° F.	24	3400	150 milliseconds

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4.2.1 *Lot.*

4.2.1.1 *Submission of product.* The product shall be submitted in accordance with Standard MIL–STD–105 or Standard MIL–STD–414, as applicable.

4.2.1.2 *Size of lot.* A lot shall consist of an identifiable group of assembled units containing metal parts assemblies produced by a homogeneous process and loaded with a single lot of cartridges.

4.2.2 *Examination.* One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein using the sampling plan and associated criteria in table III. The total number of allowable defects per class for either the major or minor classification shall not exceed the acceptance number of table III. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

TABLE III.

Lot size	Sample size	Major		Minor	
		Accept	Reject	Accept	Reject
1 – 300	100% Examination				
300 – 800	50	0	1	1	2
801 – 1800	90	1	2	2	3

4.2.2.1 Cartridge, Aircraft Canopy Remover, M29A2, Dwg. D–8593625 covering a detail of Dwg. F–8595439.

Defect	*Method of inspection*
Critical:	
1. Mixed cartridges	Visual.
2. Metal defective	Visual.
3. O ring missing	Visual
4. Sealing disc missing, damaged or improperly secured	Visual.
5. Marking missing, incorrect or illegible	Visual.
Major:	
101. O ring split, cut or cracked	Visual.
102. Foreign matter	Visual.
Minor:	
201. Case scratched, wrinkled, bulged or dented	Visual.
202. Varnish missing or inadequate	Visual.

4.2.2.2 Assembly, Dwg. F–8595439, less all Dwgs. except F–8595442, C–8595443, D–8595445, D–8595446, C–8595447, C–8595448, D–8595450, A–10521229 and CKCX2–501224.

Defect	*Method of inspection*
Critical:	
1. Gas port hole obstructed	Visual.
2. Any part missing	Visual.
3. Improper assembly	Visual.
Major:	
101. Thread damaged	Visual.
102. Burr	Visual.
103. Foreign matter	Visual.
Minor:	
None defined.	

4.2.2.3 Assembly, Dwg. F–8595439, less all Dwgs. except C–8595453, F–8595454, C–8595455, D–8595460 and D–10521228.

Defect	*Method of inspection*
Critical:	
1. Tubes not free to slide	Manual.
2. Any part missing	Visual.
3. Improper assembly	Visual.
Major:	
None defined.	
Minor:	
None defined.	

4.2.2.4 Assembly, Dwg. F–8595439, less Dwgs. C–8595457 and F–8595458.

Defect	*Method of inspection*
Critical:	
1. Boss face surface of firing pin stop not in complete contact with cartridge	Gage.
2. Any part missing	Visual.
3. Improper assembly	Visual.
Major:	
None defined.	
Minor:	
None defined.	

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4.2.2.5 Assembly, Dwg. F–8595439, less all Dwgs. except C–8595457 and F–8595458.

Defect	Method of inspection
Critical:	
1. Gasket missing or improperly assembled	Visual.
Major:	
None defined.	
Minor:	
None defined.	

4.2.2.6 Assembly, Dwg. F–8595439.

Defect	Method of inspection
Critical:	
1. Mixed removers	Visual.
2. Marking missing, incorrect or illegible	Visual.
3. Any part missing, except as otherwise classified	Visual.
4. Improper assembly	Visual.
Major:	
101. Total length, max.	Gage.
102. Length from mounting hole to face of flange	Gage.
103. Firing head not free to rotate 360 degrees	Manual.
Minor:	
201. Location of labels, incorrect	Visual.
202. Shipping plug, missing	Visual.
203. Protective finish, inadequate or defective	Visual.

4.2.2.7 Fiber Container, Dwg. C–8823715.

Defect	Method of inspection
Critical:	
None defined.	
Major:	
None defined.	
Minor:	
201. Gap between body and cover, max.	Manual.
202. Contents loose	Manual.
203. Metal ends loose or distorted	Manual.
204. Cut or damaged through all impregnated layers	Visual.
205. Marking missing, incorrect or illegible	Visual.
206. Tape incomplete or badly wrinkled	Visual.
207. Length of tear tab, min.	Visual.
208. Color of tape incorrect	Visual.

4.2.2.8 Packing Box, Dwg. C–8823716.

Defect	Method of inspection
Critical:	
None defined.	
Major:	
None defined.	
Minor:	
201. Contents loose	Manual.
202. Hardware missing, loose, broken or improperly engaged	Manual/Visual.
203. Strapping missing, loose, broken or improperly assembled	Manual/Visual.
204. Contents of box exposed	Visual.
205. Box damaged	Visual.
206. Marking missing, incorrect or illegible	Visual.
207. Number of units incorrect	Visual.
208. Car seal missing, unsealed or improperly positioned	Visual.

4.2.3 *Inspection testing.* The following tests shall be performed in accordance with 4.3, except as provided in Specification MIL–I–45208.

4.2.3.1 *Breakway torque.* The breakway torque shall be applied to each threaded joint listed in 3.2. Failure of any remover to comply with the requirements of 3.2, table I shall be cause for rejection of the remover.

4.2.3.2 *Residual magnetism.* Each remover shall be tested for excessive residual magnetism. Failure of any remover to comply with the requirement of 3.4 shall be cause for rejection of the remover.

4.2.3.3 *Function.* The following tests shall be performed by the Government:

4.2.3.3.1 *Ballistic.* Fourteen units shall be tested to determine compliance with the requirements of 3.5.1, 3.5.3 and 3.5.4. The removers shall be temperature conditioned as provided in table IV and 4.3.3.1.1. The variables sampling plan of Standard MIL–STD–414, example C–2, for the AQL listed in table IV shall be used for each sample of 10 units. Failure of any unit to comply with the requirements of 3.5.1, 3.5.3 and 3.5.4 or failure of a sample to meet the criteria of the variables sampling plan shall be cause for rejection of the lot.

4.2.3.3.2 *Locked shut.* Two removers from each lot shall be tested locked shut (see 6.3).

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Failure of any remover to comply with the requirements of 3.5.2, 3.5.3 and 3.5.4 shall be cause for rejection of the lot.

TABLE IV.

Temperature	Sample size	AQL
—65° to —70° F.	2	
65° to 75° F.	10	0.65%
160° to 165° F.	2	

4.2.4 *Inspection equipment.* The examination and tests shall be made using equipment listed on IEL–8595439, except as provided in Specification MIL–I–45208.

4.3 **Test methods and procedures.**

4.3.1 *Breakway torque.* A torque wrench, set at the torque requirements specified in 3.2, shall be used to determine the adequacy of the joint. The torque shall be applied to the firing head, cap and sleeve respectively with the unit secured in the device.

4.3.2 *Residual magnetism.* The remover shall be tested in an area free of local magnetic effects, by placing the remover 6 inches from and at the same height as the compass, in a north-south horizontal position (firing head end at south) with the compass set in an east-west heading as shown in figure 1. The remover shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 180 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass.

4.3.3 *Function.*

4.3.3.1 *Ballistics.* The test sample shall be rigidly supported by the mounting points and fired with an M3A1 Initiator connected to the remover with 10 feet of aircraft hose, size 4, Standard MS–28741. The sample shall be tested within 10 minutes after removal from the conditioning chamber. When this time limit is exceeded, the sample shall be reconditioned at the original temperature.

4.3.3.1.1 *Temperature conditioning.* A standard M4 Remover, loaded with a dummy M29A2 Cartridge, containing a temperature sensitive element shall be placed in the conditioning chamber at the same time as the samples. The conditioning time shall be a minimum of one hour longer than the time necessary to bring the element contained in the dummy cartridge to the required temperature.

4.3.3.2 *Locked shut.* Two removers shall be temperature conditioned in accordance with 4.3.3.1.1 and as follows: one remover at 160 degrees F. to 165 degrees F., and the other at minus 65 degrees F. to minus 70 degrees F. The test samples shall be rigidly supported at the mounting points and the tubes shall be restrained from moving. The test samples shall be fired with an M3A1 Initiator connected to the remover with 10 feet of aircraft hose, size 4, Standard MS–28741.

5. PREPARATION FOR DELIVERY

5.1 **Packing and marking.** Level A—Packing and marking shall be in accordance with Drawings C–8823715 and C–8823716.

5.2 **Data cards.** Data card information shall be as specified in Specification MIL–A–2550.

6. NOTES

6.1 A mechanical failure is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.2 Ignition delay is defined as the time that elapses between the firing of the cartridge primer and the beginning of the continuous rise on the thrust time record.

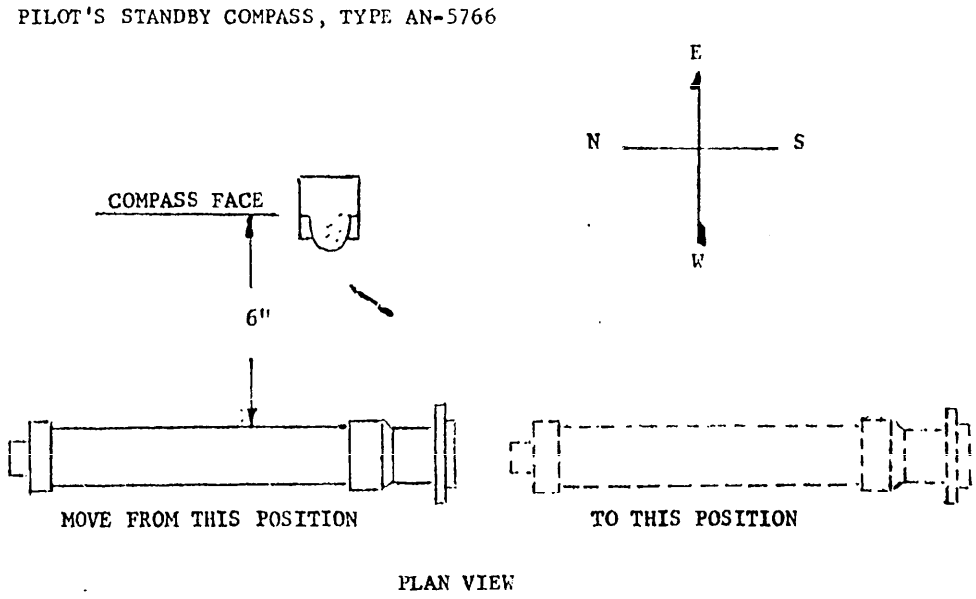
6.3 Locked shut is defined as a condition in which the remover tubes are restrained from moving upon firing of the remover.

6.4 Velocity applies at tube separation.

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NOTE:
REMOVER AND COMPASS SHALL LIE IN A HORIZONTAL PLANE

FIGURE 1. *Layout for determination of residual magnetism in remover, aircraft canopy M4.*

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SPECIFICATION ANALYSIS SHEET

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INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

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FORM 1426
APR 63

REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

PLATE NO. 15419

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209. Diameter of counterbore, min.

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